Quiz practice.

Q1.

$$x_1 + 2x_2 + x_3 = 4$$
$$2x_1 + 4x_2 + 3x_3 + x_4 = 9$$
$$x_1 + 2x_2 + 2x_3 + x_4 = 5$$

① Augmented matrix:

$$\left[\begin{array}{cccc|c} 1 & 2 & 1 & 0 & 4 \\ 2 & 4 & 3 & 1 & 9 \\ 1 & 2 & 2 & 1 & 5 \end{array}\right]$$

② Row Reduction

$R_2 \leftarrow R_2 - 2R_1 \Rightarrow$

$2 - 2(1) = 0$
$4 - 2(2) = 0$
$3 - 1(2) = 1$
$1 - 0(2) = 1$
$9 - 4(2) = 1$

$$\left[\begin{array}{cccc|c} 1 & 2 & 1 & 0 & 4 \\ 0 & 0 & 1 & 1 & 1 \\ 1 & 2 & 2 & 1 & 5 \end{array}\right]$$

$R_3 \leftarrow R_3 - R_1 \Rightarrow$

$$\left[\begin{array}{cccc|c} 1 & 2 & 1 & 0 & 4 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{array}\right]$$

$R_3 \leftarrow R_3 - R_2 \Rightarrow$

$$\left[\begin{array}{cccc|c} 1 & 2 & 1 & 0 & 4 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

Convert RREF

$R_1 \leftarrow R_1 - R_2 \Rightarrow$

$$\left[\begin{array}{cccc|c} 1 & 2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

③ Basic variables: x_1, x_3
Free variables: x_2, x_4

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④ $x_1 + 2x_2 - x_3 = 3$

$x_3 + x_4 = 1$

Let $x_2 = s$ $\quad x_3 = 1 - t$

$x_4 = t$ $\quad x_1 = 3 - 2s + t$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 3 - 2s + t \\ s \\ 1 - t \\ t \end{bmatrix}$$

$$\begin{matrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{matrix} = s[\quad] + t[\quad] + 4$$

④ $$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 1 \\ 0 \end{bmatrix} + s\begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t\begin{bmatrix} 1 \\ 0 \\ -1 \\ 1 \end{bmatrix}$$

where $s, t \in k$

∞ many solutions

Q2.

$$v_1 = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} \quad v_2 = \begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix} \quad v_3 = \begin{bmatrix} 1 \\ 1 \\ h \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 5 & 1 \\ -1 & 1 & h \end{bmatrix}$$

1) $Ax = 0$ or $x_1 v_1 + x_2 v_2 + x_3 v_3 = 0$

2) Row

$R_2 \leftarrow$

R_3

R_3

R_0

for

b

w

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2) Row Reduction.

$R_2 \leftarrow R_2 - 2R_1$

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & -1 \\ -1 & 1 & h \end{bmatrix}$$

$R_3 \leftarrow R_3 + R_1$

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & -1 \\ 0 & 3 & h+1 \end{bmatrix}$$

$R_3 \leftarrow R_3 - 3R_2$

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & h+4 \end{bmatrix}$$

No free variable → only trivial solution → → linearly independent

free variable → dependent → non trivial solution exists.

for a linear dependence we need:

$h + 4 = 0$

$h = -4$

with $h = -4$ → $x_1 + 2x_2 + x_3 = 0$

$x_2 - x_3 = 0$

choose $x_3 = 1$ then $x_2 = +1$

$x_1 = -3$

$x_1 = -3 \quad x_2 = 1 \quad x_3 = 1$

Result: $-3v_1 + v_2 + v_3$

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Q3

~~k~~ $k^2 \to k^2$

rotating 90° counterclockwise

$$(x, y) \to (-y, x)$$

reflect through the x-axis

$$(x, y) \to (x, -y)$$

a) standard basis vectors are

$$e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \qquad e_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Rotated $e_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ $\qquad e_2 = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$

reflected through x-axis $T(e_1) = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$

$T(e_2) = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$

$$A = [T(e_1)\ T(e_2)]$$

$$A = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$

b) Compute $T(3, 2)$

$$T(3,2) = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} = (-2, -3)$$

$0(3) + (-1)(2) = -2$

$-1(3) + 0(2) = -3$

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c) $A = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$

$R_1 \leftrightarrow R_2 \rightarrow \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

$R_1 \leftarrow -R_1 \rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$R_2 \leftarrow -R_2$

pivots are in every row, column, T is onto, T is one-to-one

for 2x2 we can also determine whether it is onto by finding its det(A)

$\det(A) \neq 0 \rightarrow$ onto

$\det(A) = (0)(0) - (-1)(-1) = \underline{-1}$ T=onto

Q4. LU Factorization

$$\begin{bmatrix} \textcircled{2} & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{bmatrix} = A$$

Find L and U

① $m_{21} = \frac{m_{21}}{m_{11}}$ $\quad m_{21} = \frac{4}{2} = 2$

$R_2 \leftarrow R_2 - 2R_1 \Rightarrow \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 8 & 7 & 9 \end{bmatrix}$

② $m_{31} = \frac{m_{31}}{m_{11}}$ $\quad m_{31} = \frac{8}{2} = 4$

$R_3 \leftarrow R_3 - 4R_1 \Rightarrow \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 3 & 5 \end{bmatrix}$

③ $m_{32} = \frac{m_{32}}{m_{22}}$ $\quad m_{32} = \frac{3}{1} = 3$

$R_3 \leftarrow R_3 - 3R_2 \Rightarrow \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix} = U$

$m_{21} = 2$
$m_{31} = 4$
$m_{32} = 3$

$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 3 & 1 \end{bmatrix} = L$

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Q 5.

$Ax = b$ with $b = \begin{bmatrix} 3 \\ 7 \\ 13 \end{bmatrix}$

do it in two steps:

1) $Ly = b$ by forward substitution
2) $Ux = y$ by backward substitution.

1) $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 3 & 1 \end{bmatrix} \cdot \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 7 \\ 13 \end{bmatrix}$

L

$y_1 = 3$

$2y_1 + y_2 = 7 \rightarrow y_2 = 1$

$4y_1 + 3y_2 + y_3 = 13 \rightarrow y_3 = -2$

$y = \begin{bmatrix} 3 \\ 1 \\ -2 \end{bmatrix}$

2) $\begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ -2 \end{bmatrix}$

U

$x_3 = -1$

$x_2 + x_3 = 1 \rightarrow x_2 = 2$

$2x_1 + x_2 + x_3 = 3 \rightarrow x_1 = 1$

$x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$

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$$y = \begin{bmatrix} 3 \\ 1 \\ -2 \end{bmatrix} \qquad x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

$$Ax = b$$

$$\begin{bmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{bmatrix} \begin{bmatrix} x_1 = 1 \\ x_2 = 2 \\ x_3 = -1 \end{bmatrix} = \begin{bmatrix} 3 \\ 7 \\ 13 \end{bmatrix}$$

Q6. Determinants.

$R_i \leftarrow k_i + kR_j$ No change

$R_i \leftarrow > R_j$ Swaps sign

$R_i \leftarrow kR_i$ $k \cdot \det(A)$

$$\begin{bmatrix} 1 & 2 & 0 & 3 \\ 2 & 5 & 1 & 7 \\ 0 & 1 & 3 & 4 \\ 1 & 3 & 3 & 9 \end{bmatrix}$$

② $m_{21} = \frac{m_{21}}{m_{11}} = \frac{2}{1} = 2$

$R_2 \leftarrow R_2 - 2R_1 \Rightarrow$

$$\begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 3 & 4 \\ 1 & 3 & 3 & 9 \end{bmatrix}$$

② $m_{32} = \frac{m_{32}}{m_{22}} = \frac{1}{1} = 1$

$R_3 \leftarrow R_3 - 1 \cdot R_2 \Rightarrow$

$$\begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 2 & 3 \\ 1 & 3 & 3 & 9 \end{bmatrix}$$

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③ $m_{41} = \frac{m_{41}}{m_{11}} = \frac{1}{1}$

$R_4 \leftarrow R_4 - R_1 \Rightarrow \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 2 & 3 \\ 0 & 1 & 3 & 6 \end{bmatrix}$

④ $m_{42} = \frac{m_{42}}{m_{22}} = \frac{1}{1}$

$R_4 \leftarrow R_4 - R_2 \Rightarrow \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 2 & 3 \\ 0 & 0 & 2 & 5 \end{bmatrix}$

⑤ $m_{43} = \frac{m_{43}}{m_{33}} = \frac{2}{2} = 1$

$R_4 \leftarrow R_4 - 1 \cdot R_3 \rightarrow \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 2 & 3 \\ 0 & 0 & 0 & 2 \end{bmatrix} = U_\Delta$

$$\begin{aligned} \det(A) &= u_{11} \cdot u_{12} \cdot u_{13} \cdot u_{14} \\ &= 1 \cdot 1 \cdot 2 \cdot 2 \\ &= 4 \end{aligned}$$

All five operations are row replacements, so the determinant remains unchanged.

Q7.

3×3 matrices

$\det(A) = -2$

$\det(B) = 5$

(a) $\det(2A) = 2^3 \cdot (-2) = 8 \cdot (-2) = -16$

(b) $\det(A^{-1}) = \frac{1}{-2} = -\frac{1}{2}$

(c) $\det(A^T B) = -2 \cdot 5 = -10$

(d) $\det(A^3) = (-2)^3 = -8$

(e) swapping two rows : 1) 2

multiplying one row by 3 : 2|3 · 2 = 6

$\det(2A)$ = scalar Multiplication property

$\det(A^{-1})$ = Inverse determinant property

$\det(A^T B)$ = transpose property and Product property

$\det(A^3)$ = power property

swapping rows = Row Interchange property

multiplying one row by 3 = Row scaling property

Q. 8.

a)

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 2 & 5 & 7 \end{bmatrix} \qquad [A|B] = \left[\begin{array}{ccc|c} 1 & 2 & 3 & 3 \\ 0 & 1 & 1 & 1 \\ 2 & 5 & 7 & 7 \end{array}\right]$$

$R_3 \leftarrow R_3 - 2R_1 \Rightarrow$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 3 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{array}\right]$$

$R_3 \leftarrow R_3 - R_2 \Rightarrow$ ⟶

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$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 3 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

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inconsistent if we got $[0\ 0\ 0 \mid 5]$

$[0\ 0\ 0 \mid 0]$ its fine

$Ax=b$ is consistent

$$A|b = \begin{bmatrix} 1 & 2 & 3 & | & 3 \\ 0 & 1 & 1 & | & 1 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

Convert to RREF

$$R_1 \leftarrow R_1 - 2R_2 \Rightarrow \begin{bmatrix} 1 & 0 & 1 & | & 1 \\ 0 & 1 & 1 & | & 1 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

basic : x_1, x_2 $\quad x_1 + x_3 = 1$

free : x_3 $\quad x_2 + x_3 = 1$

let $x_3 = t$ $\quad x_1 = -t + 1 = 1 - t$

$x_2 = 1 - t$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = t\begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad t \in R$$

b is linear combination of the columns of A

$\Leftrightarrow$

$Ax=b$ is consistent.

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$$\begin{bmatrix} 1 & 2 & 3 & 3 \\ 0 & 1 & 1 & 1 \\ 2 & 5 & 7 & 8 \end{bmatrix}$$

Row Reduction

$R_3 \leftarrow R_3 - 2R_1 = \begin{bmatrix} 1 & 2 & 3 & 3 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 2 \end{bmatrix}$

$R_3 \leftarrow R_3 - R_2 \quad \begin{bmatrix} 1 & 2 & 3 & 3 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

last row means

$0X_1 + 0X_2 + 0X_3 = 1$

$0 = 1$ not

$Ax = b$ is inconsistent

$b(3, 1, 1)$ is NOT a linear combination of the columns of A.

e) Does $Ax = b$ have a solution for every b in $\mathbb{R}^3$?
Is T onto?
For every b we need a pivot in every row of A

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$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 2 & 5 & 7 \end{bmatrix}$$

$R_3 \leftarrow R_3 - 2R_1 \rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$

$R_3 \leftarrow R_3 - R_2 \rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

$R_1 \leftarrow R_1 - 2R_2 \rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

only two pivots

solution for every b in $\mathbb{R}^3$ → 3 pivots ✗
2 pivots → NO

A is not onto

A is not one-to-one

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Q.9. Permutation Matrix.

$$\begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 1 \\ 2 & 7 & 9 \end{bmatrix}$$

a) Swap two rows, find the permutation matrix P, and calculate PA
b) Find L and U such that PA = LU
c) Use PA = LU to calculate det(A)

a) $a_{11} = 0$ need nonzero pivot

$R_1 \leftrightarrow R_2$ $\quad PA = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 2 & 7 & 9 \end{bmatrix}$

Find P:

Identity matrix:

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

b) $PA = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 2 & 7 & 9 \end{bmatrix}$

1) $m_{31} = \frac{m_{31}}{m_{11}} = \frac{2}{1} = 2$

$R_2 \leftarrow R_2 - 2R_1$

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 0 & 3 & 7 \end{bmatrix}$$

$m_{32} = \frac{m_{32}}{m_{22}} = \frac{3}{1} = 3$

2) $m_{32} = m_{22}$

$R_3 \leftarrow R_3 - 3R_2$ $\rightarrow$ $\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} = U$

$m_{31} = 2$

$m_{32} = 3$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 3 & 1 \end{bmatrix}$$

c) Calculate det(A)

$PA = LU$

$$\det(P)\det(A) = \det(L)\det(U)$$

swapping two rows:

$$\det(P) = -1$$

$\det(L) = 1$

$\det(U) = 1$

$$-1 \cdot \det(A) = 1 \cdot 1$$

$$\boxed{\det(A) = -1}$$

Q.10. Cofactor Expansion

Cofactor Expansion along the first row:

$$\det(A) = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$$

$$C_{ij} = (-1)^{i+j} \cdot M_{ij}$$

$$\begin{matrix} + & - & + \\ - & + & - \\ + & - & + \end{matrix}$$

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Cramer's Rule:

if $\det(A) \neq 0$:

$$x = \frac{\det(A_x)}{\det(A)} \qquad y = \frac{\det(A_y)}{\det(A)}$$

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 1 & 3 & -1 \\ 0 & 2 & 4 \end{bmatrix} \qquad B = \begin{bmatrix} 1 \\ 8 \\ 0 \end{bmatrix}$$

a) $\det(A) = ?$ using cofactor expansion along 1st row

$$\det(A) = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$$

1) $C_{11} = (-1)^2 \cdot M_{11} = 1 \cdot 14 = 14$

$M_{11} = 3 \cdot 4 - 2 \cdot (-1) = 14$

2) $C_{12} = (-1)^3 \cdot M_{12} = -1 \cdot 4 = -4$

$M_{12} = 1 \cdot 4 + 0 \cdot (-1) = 4$

3) $M_{13} = 1 \cdot 2 - 3 \cdot 0 = 2$

$C_{13} = (-1)^4 \cdot 2 = 1 \cdot 2 = 2.$

$\det(A) = 2 \cdot 14 + 0 \cdot (-4) + 1 \cdot 2 = 30$

b) $x_2 = \dfrac{\det(A_{x_2})}{\det(A)}$ $\qquad A_{x_2} = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 8 & -1 \\ 0 & 0 & 4 \end{bmatrix}$

Using Sarrus's Rule $\begin{bmatrix} 2 & 1 & 1 \\ 1 & 8 & -1 \\ 0 & 0 & 4 \end{bmatrix} \begin{matrix} 2 & 1 \\ 1 & 8 \\ 0 & 0 \end{matrix}$

$\det(A_{x_2}) = 2 \cdot 8 \cdot 4 + 0 + 0 - 0 - 4 + 0 = 60$

$x_2 = \dfrac{60}{30}$

$x_2 = \boxed{2}$

$\det(A_{x_2}) = a_{31}C_{31} + a_{32}C_{32} + a_{33}C_{33}$

c) A is invertible since $\det(A) \neq 0$